THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR JOINT EXAMINATION FOR GOVERNMENT SCHOOLS –JEGOS

PRE-MOCK EXAMINATION
CHEMISTRY 1
MARKING SCHEME 2024

SECTION A (16 marks)

1. Multiple choice (10marks @ 01mark)

QNS	i	ii	iii	iv	v	vi	vii	viii	ix	x
ANS	A	A	D	E	D	D	B	D	A	E

2. Match the items (06marks @ 01mark)

LIST A	i	ii	iii	iv	v	vi
LIST B	D	H	E	A	C	G

SECTION B (54marks)

3. (a) (i) Would use the process of decantation by letting the water settle for a day, or filtration using a piece of cloth (1½ @ = 03 marks)
(ii) Would then boil the water, or treat with chemical purifiers (example water guard…)

(b) Tests for water
- Water can be recognized by its action of turning white anhydrous copper (II) sulphate to blue.
- The presence of water can also be shown by the use of cobalt chloride paper. This is a filter paper impregnated with cobalt (II) chloride. The paper is blue in colour. The blue paper turns pink when in contact with water.
- If its pure water, the boiling point is exactly 100ºC and freezes at 0ºC at pressure of one atmosphere (760 mmHg). (2 @ = 06 marks)

4. (a)

$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_2O_{(l)} + CO_{2(g)}$ (1½ marks)

(b) 2 grams of calcium carbonate were reacted with excess dilute hydrochloric acid and gas Y was liberated.

From the balanced equation

1 mole $CaCO_{3(s)}$ = 1 mole of $CO_{2(g)}$ = 22.4 dm^3 (01 mark)

2 grams = moles?

Moles = mass/molar mas

= 2/100 (1½ marks)

= 0.02 moles

So, 1 mole $CaCO_{3(s)}$ gives 1 mole $CO_{2(g)}$

0.02 moles $CaCO_{3(s)}$ gives 0.02 moles $CO_{2(g)}$ (1½ marks)

And

1 mole $CO_{2(g)}$ = 22.4 dm^3

0.02 moles $CO_{2(g)}$ = ?

$0.02 \times 22.4\ dm^3 = 0.448\ dm^3$ (1½ marks)

Volume og gas y produced 0.448 dm^3

(c)

Mass = Moles × Molar mass

Molar mass of CO_2 = 44

= 0.02 × 44 (1½ marks)

= 0.88g

Mass of gas Y produced = 0.88g (00½ mark)

5. (a)

(i) The particles/molecules gain energy and vibrate more violently and freer

(ii) The loosely held water molecules lose energy and become more stationary and compacted

(iii)The gas molecules are completely free of each other and therefore gas can assume a large volume, which can be compressed to a much smaller volume since the molecules are very small

(iv)The heat from the sun changes the water into vapour by increasing the energy of the water molecules. (1½@ = 06 marks)

(b) (i) The zinc and copper (Brass is an alloy of copper and zinc) can be separated by physical means such as melting while the constiuents water cannot be separated by physical means

(ii). The ratio of the components of brass can vary, while the ratio of the components of water ae always constant. (1½@ = 03 marks)

6. (a) Differences between oxygen and hydrogen ANY THREE (03marks @ 01mark)

S/N	HYDROYGEN	OXYGEN
(i)	It burn on air	It does not burn but support combustion
(ii)	Less dense than air	Dense than air
(iii)	Is not used for helping patient with difficult breathing	Used for helping patient with difficult breathing
(iv)	It is reducing agent	It is oxidizing agent

(b) With the aid of chemical equations, briefly describe the following process.

(i) The removal of temporary hardness of water by **Boiling**

Temporary hard water softens by boiling to promote formation of carbonate from hydrogen carbonate. Precipitate (substance to be deposited in solid form from a

solution) of carbonate filtered and leaves water soft (01.5marks)

$Ca(HCO_3)_2(aq) \longrightarrow H_2O\ (l) + CO_2\ (g) + CaCO_3\ (s)$ (01.5marks)

(ii) The removal of permanent hardness of water by addition of washing soda Sodium carbonate is added to precipitate calcium carbonate Which are removed by filtration or decantation. (01.5marks)

$CaSO_4(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s) + Na_2SO4(aq)$ (01.5marks)

7. (a)

(i) Production of sulphur dioxide, Purification of sulphur dioxide, Catalytic conversion of sulphur dioxide to sulphur trioxide, and conversion of sulphur trioxide to sulphuric.

0½@ = 02 marks

(ii) $S_{(s)} + O_{2(g)} \rightarrow SO_{2(g)}$ (burning sulphur in the air)

$4FeS_{2(s)} + 11O_{2(g)} \rightarrow 8SO_{2(g)} + 2Fe_2O_{3(s)}$ (Burning sulphide ores…)

$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2SO_{2(g)} + 2ZnO_{(s)}$ (Burning sulphide ores…) (*Any one* 01 mark)

(iii) The gas mixture is passed through concentrated sulphuric acid to remove all moisture. 01 mark

(iv) Arsenic acid (As_2O_3), dust, and moisture 0½@ = 01½ marks

(v) If not removed, will poison the catalyst thereby rendering it useless 0½ mark

(vi) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons SO_{3(g)} + \text{heat}$ (V_2O_5) 01 mark

(vii) $H_2SO_{4(aq)} + SO_{3(g))} \rightarrow H_2SO_{7(g)}$ 01 mark

(viii) $H_2SO_{7(g)} + H_2O_{(l)} \rightarrow 2H_2SO_{4(aq)}$ 01 mark

8. (a).When exposed to open air, ethanol is oxidized to ethanoic acid. The reaction for the process occurs thus: 01½ marks

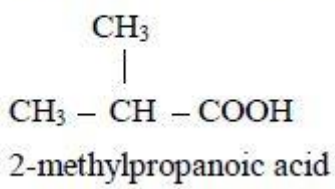
$C_2H_5OH_{(l)} + 2[O]_{(g)} \rightarrow CH_3COOH_{(l)} + H_2O_{(l)}$ 01½ marks

(b)

- Butanoic acid, C_3H_7COOH or $C_3H_7CO_2H$ or $CH_3CH_2CH_2COOH$

Isomers:

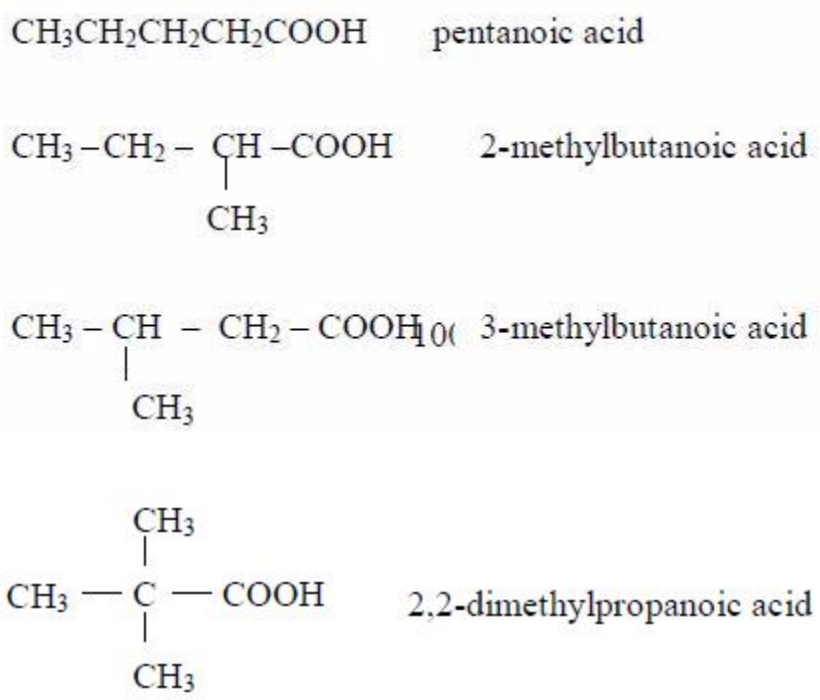
$CH_3CH_2CH_2COOH$ butanoic acid 01 mark



2-methylpropanoic acid 01 marks

- Pentanoic acid, C_4H_9COOH

Isomers



(01 @ = 04 marks)

SECTION C (30 marks)

9. Mining is the extraction of valuable geological materials and minerals from the surface of the Earth. Extraction consists of separating the metal from minerals in the ore. (02 marks)

   Mining can cause the following environment problems:

   - Land subsidence (sagging): Holes created due to underground mining cause land to sink (or subside). This is because the holes underneath the ground cause imbalance in weight of the soil above the ground. This may result to severe damage to buildings and other infrastructures such as roads, railway trucks and so forth. (any six 02@ = 12 marks)
   - Poisonous substances: Poisonous compounds (for example of lead, cadmium and arsenic) are found in many ores. These may be washed into the soil and streams because of the mining process. If they happen to reach the water, they can kill fish and plant life, and can end up in your food …
   - Noise and dust: Mining activities produce a lot of noise and dust. Noise and dust can be caused by haulage trucks, rock blasting and crushing, drilling operations and heavy traffic. Everything for miles around the mine may get covered with dust.
   - Great heaps of earth material: unwanted rock material, after the metal has been extracted from the ore, gets heaped up in tips. These are unsightly. They can be unstable and therefore dangerous. During heavy rains, a landslide is likely to occur, a catastrophe that often results to loss of life and destruction of property.
   - Soil erosion: Before mining operation is carried out, the natural vegetation on or around the mining site is usually cleared up in order to give enough room to mining activities. The consequent removal of vegetation cover leaves the soil bare and, therefore, susceptible to erosion.
   - Air pollution: Large-scale mineral extraction results to production of gases such as sulphur dioxide, carbon dioxide and other bad gases which are emitted to the atmosphere. These gases may bring about a green house effects and even cause acid rains.

Conclusion: Although Mining and mineral extraction is important for economic development and general human welfare, it can cause many environment problems as explained above. Therefore, it should be carried out with great check and balances such as by incorporating intervention measures.
(01 mark)

Total: 15 marks

10.

Applications of acid-base neutralization in everyday life

Indigestion and pain relief

Excess acid in the stomach causes indigestion, which can be painful; we take an anti-acid to neutralize the effects of acid indigestion.

An example of an anti-acid used treat indigestion and pain is insoluble magnesium hydroxide, $Mg(OH)_2$. The neutralization reaction equation is:

$$Mg(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$

Insect stings treatment

The bee sting is acidic in nature. It contains formic acid (methanoic acid); it can be neutralized by rubbing on *calamine solution*, which contains zinc carbonate or baking soda (sodium hydrogencarbonate).

$$HCOOH_{(aq)} + NaHCO_{3(aq)} \rightarrow HCOONa_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$

Wasp stings on the other hand are alkaline in nature, and can be neutralized with vinegar (ethanoic acid).

Descaling kettles

The limescale ($CaCO_3$) is formed inside kettles and water heaters when hard water is boiled. The limescale can be removed by treatment with an acid that is strong enough to react with $CaCO_3$, but not strong enough to damage the metal. Vinegar can be used to discale kettles:

$$2CH_3COOH_{(aq)} + CaCO_3 \rightarrow Ca(C_2H_3O_2)_{2\ (aq)} + H_2O_{(l)} + CO_{2(g)}$$

Prevention of tooth decay

Food remnants (especially sugary food) in the mouth are acted upon by bacteria to form acids. The acid attacks the tooth enamel and cause tooth decay. Tooth decay begins when the pH falls below 5.8

To help prevent tooth decay many types of toothpaste contain basic substances to neutralize the acids produced by these bacteria in the mouth.

(Any five 03@ = 15marks)

Soil treatment

Most plants grow best when pH of the soil is between 6.5 and 7.0. If the soil pH is below 6.0, the soil is too acidic. Above the pH of 8.0, the soil is too alkaline.

If the soil is too acidic, liming is done to adjust its pH. Liming means addition of quicklime (calcium oxide), slaked lime (calcium hydroxide) or powdered chalk or limestone (calcium carbonate) to an acidic soil. These compounds (bases) have the effect of neutralizing the acidity of the soil.

Factory wastes treatment

Liquid wastes from factories often contain acid. If it reaches a river, lake or ocean, the acid will kill fish and other aquatic life. This can be prevented by adding slaked lime (calcium hydroxide) to the waste, to neutralize the acid before being dumped into water bodies. **Total: 15 marks**

11.

. (i) Volume of chlorine gas = 100 liters

Molar mass of chlorine gas is = 71g/mol

71g of chlorine gas occupy 22.4 dm^3 at STP

xg of chlorine gas occupy 100 dm^3 at STP (01 mark)

The 100 litres of chlorine is converted in to mass/ expressed in terms of mass

$\frac{100\ x\ 71}{22.4}$ = 316.96g of Chlorine gas at STP (01½ mark) (01 marks)

$\frac{\text{Ar.I.t.}}{\text{V.F.}}$ = 316.96 g (02 marks)

Where I = current (Unknown), V = valence of chlorine (2239) , F = 96,500 coulombs, Ar= relative atomic mass of chlorine gas (71g) and t = time (3,600 seconds) (02 marks)

$\frac{316.96\ x\ V\ x\ F}{Ar.t} = I$ (02 marks)

$\frac{316.96\ x\ 2\ x\ 96500}{71\ x\ 3600} = I$ (01 marks)

I= 239.33A (01 marks)

(ii) At the anode: $2Cl^- \rightarrow Cl_{2(g)} + 2e^-$ (01 mark)

At the cathode: $2H^+ + 2e^- \rightarrow H_{2(g)}$ (01 mark)

(iii) Sodium hydroxide: it forms as hydrogen gas is liberated at the cathode causing water to dissociate into OH^- and hydrogens, (01½ mark)

$H_2O \Leftrightarrow H^+ + OH^-$

$\downarrow$

$2H^+ + 2e^- \rightarrow H_{2(g)}$

Total: 15 marks